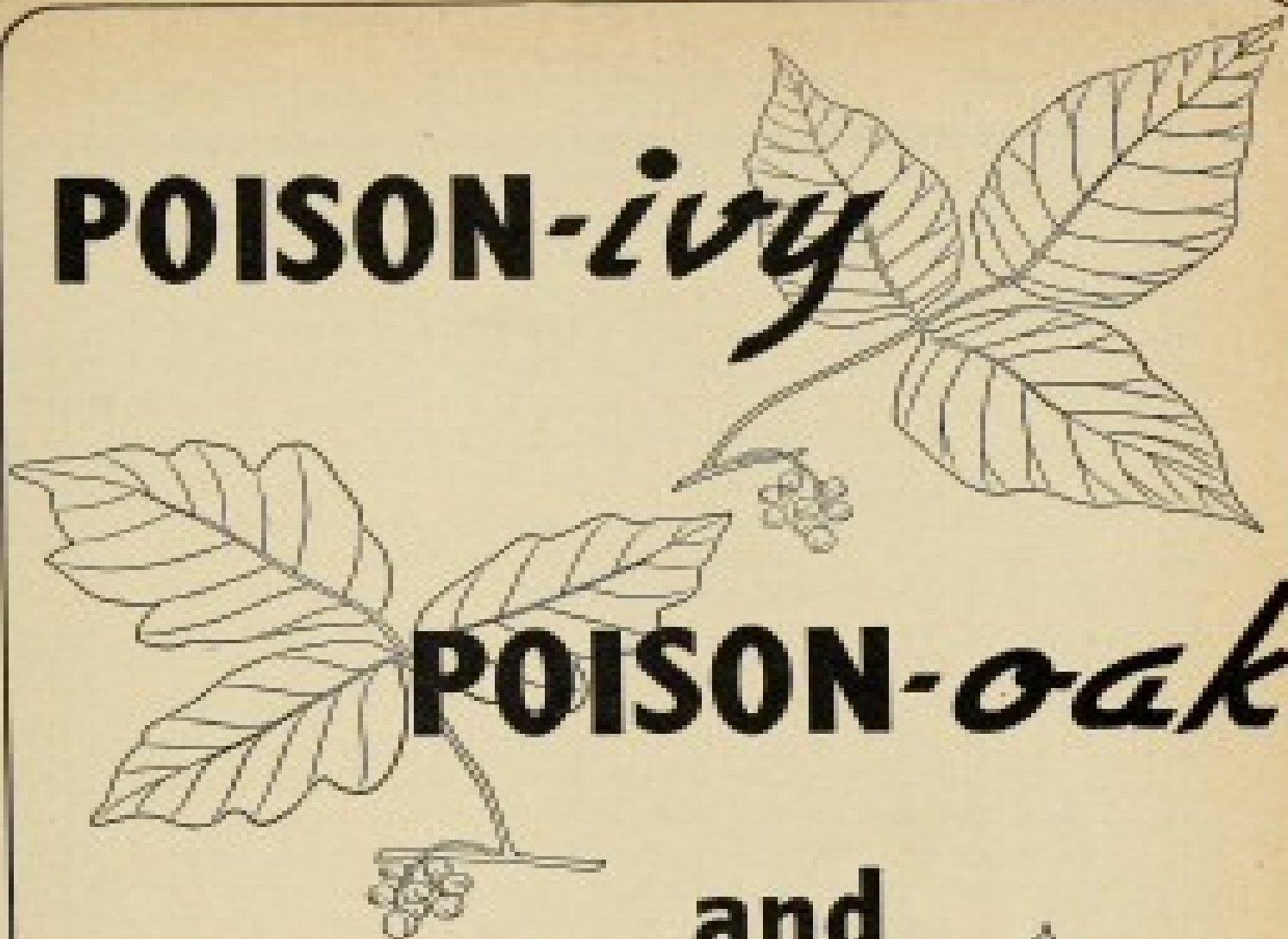


POISON-*ivy*



POISON-*oak*

and

POISON



sumac

IDENTIFICATION
PRECAUTIONS
ERADICATION

•

Farmers' Bulletin No. 1972

U. S. DEPARTMENT OF AGRICULTURE

POISON-*ivy*

POISON-*oak*

and

POISON *sumac*

IDENTIFICATION
PRECAUTIONS
ERADICATION
•

Farmers' Bulletin No. 1972

U. S. DEPARTMENT OF AGRICULTURE

The Project Gutenberg eBook of Poison-ivy, poison-oak and poison sumac

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Poison-ivy, poison-oak and poison sumac
Identification, precautions, eradication

Author: D. M. Crooks
L. W. Kephart

Release date: May 19, 2019 [eBook #59555]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59555

Credits: Produced by Tom Cosmas compiled from images made available
by The Internet Archive.

*** START OF THE PROJECT GUTENBERG EBOOK POISON-IVY,
POISON-OAK AND POISON SUMAC ***

POISON-ivy

POISON-*oak*

and

POISON *sumac*

IDENTIFICATION

PRECAUTIONS

ERADICATION

Farmers' Bulletin No. 1972
U. S. DEPARTMENT OF AGRICULTURE

POISON-IVY, poison-oak, and poison sumac remind most people of painful experiences to be avoided, yet many do not know any one of the offending plants or their equally poisonous relatives. Learning to recognize them on sight is relatively easy, especially by examining the distinctive identifying characters described in the pictures and legends of this bulletin. There is then a good chance to avoid them or, if one must work among them, to take preventive measures to escape poisoning.

These plants are very common throughout the United States. They are found in fields and woods, along fence rows, rock walls, and hedges, in lawns and gardens, and even sometimes vining on houses. Usually in any one locality it is necessary to be certain of the identity of only two or three of them. Maps show at a glance where they are likely to be found. Frequent observation and recognition of the plants as they are encountered almost daily is the best way to become poison-oak or poison-ivy conscious. Nonpoisonous sumacs are easily distinguished from the poisonous species by the seed heads and leaves.

The old proverb, "an ounce of prevention is worth a pound of cure," is good advice to everyone, particularly as regards ivy-poisoning. Previous escape is not proof of immunity when conditions are right, and after poisoning occurs there is no quick cure known. Some relief may be obtained and recovery

hastened by use of some of the tested remedies. In case of severe poisoning a physician should be consulted.

Poison-ivy and other poisonous plants growing in grounds frequented by people should be eradicated. In some places this can be done by careful grubbing. In others weed-killing chemicals may be better and more certainly would avert poisoning for most of us.

This bulletin supersedes Farmers' Bulletin 1166, Poison Ivy and Poison Sumac and Their Eradication.

Washington, D. C.

Issued August 1945

POISON-IVY, POISON-OAK, AND POISON SUMAC: IDENTIFICATION, PRECAUTIONS, ERADICATION

**By DONALD M. CROOKS, *principal horticulturist,
Division of Tobacco, Medicinal, and Special Crops,*
and LEONARD W. KEPHART, *senior agronomist,
Division of Cereal Crops and Diseases, Bureau of
Plant Industry, Soils and Agricultural Engineering,
Agricultural Research Administration***

Contents

	Page
A widespread menace to health	<u>1</u>
Forms of poison-ivy and poison-oak	<u>2</u>
Common poison-ivy	<u>2</u>
Oakleaf poison-ivy	<u>9</u>
Western poison-oak	<u>9</u>
Poison sumac	<u>16</u>
Confusion of poisonous and harmless sumacs	<u>19</u>
Introduced poisonous sumac and related species	<u>21</u>
Poisoning	<u>21</u>
Precautions against poisoning	<u>22</u>
Treatment for poisoning	<u>23</u>
Eradication of plants	<u>25</u>
Chemical weed killers	<u>25</u>
Eradication by mechanical means	<u>25</u>

A WIDESPREAD MENACE TO HEALTH

EVERY year sees many adults, and especially children, accidentally poisoned from contact with plants that they did not know were harmful. Had the sufferers known how to detect poisonous plants they could have avoided them and escaped the painful experience of severe skin inflammations and water blisters. Very few persons have a sufficient degree of immunity to protect them from poisonous plants, and many do not recognize them in the various forms that they assume in different parts of the country.

One or more kinds of poison-ivy—common poison-ivy, oakleaf poison-ivy, and western poison-oak—occur in abundance in almost every part of the United States. Poison sumac is of more limited distribution and occurs chiefly east of the Mississippi River, usually in swampy regions.

Poisoning by these plants is largely preventable. A knowledge sufficient to identify them in their various forms is easily gained by anyone who will make a study of pictures and general descriptions and train himself by a little diligent practice to observe the plants in his locality. It is important to become poison-oak or poison-ivy conscious. This can be accomplished only by learning to know the plants by repeated experience in observing them in their various forms. Persons who learn both to know and to avoid the plants can save themselves much inconvenience and discomfort. The simple means of learning to recognize them will pay big dividends. It is the first step to the prevention of poisoning and to the eradication of the pests from private grounds and public places.

The most effective method of prevention is to eradicate the plants, and wherever possible this should be done. They should not be allowed to grow in lawns or gardens, with ornamental shrubbery, or on houses. Communities should not countenance the growth of poison-ivy or poison-oak in school grounds, on public roadways, or in parks, especially those frequented by children. The cost of medical aid for one season in treating pupils using an ivy-infested schoolyard is likely to be more than that of eradicating the plants, to say nothing of the suffering and inconvenience they may cause.

FORMS OF POISON-IVY AND POISON-OAK

Poison-ivy and poison-oak are known by a number of local names, and several different kinds of plants are called by these names. The plants exhibit a good deal of variation throughout the United States, so much so that even technical botanists cannot agree on the number of species and names that should be applied to them. They grow in the form of (1) woody vines attached to trees or objects for support, (2) trailing shrubs mostly on the ground, or (3) erect woody shrubs entirely without support. They may flourish in the deep woods, where soil moisture is plentiful, or they may be found in very dry soil on the most exposed hillsides. The plants are most frequently abundant along old fence rows and edges of paths and roadways, and they ramble over rock walls and climb posts or trees to considerable heights. They often grow with other shrubs or vines in such ways as to escape notice.

The leaf forms among plants or even on the same plant are as variable as the habit of growth; however, the leaves always consist of three leaflets. The old saying, "Leaflets three, let it be," is a reminder of this consistent leaf character but may lead to undue suspicion of some harmless plant. Only one three-parted leaf leads off from each node on the stem. The leaves never occur in pairs along the stem. The wide range in habit of growth of plants and leaf types may be noted from a glance at the several pictures of plants and leaves.

The flowers and fruit are always in clusters on slender stems that originate in the axis of the leaves along the side of the smaller branches. The fruits usually have a white, waxy appearance and ordinarily are not hairy, but may be so in some forms. The plants do not always flower and bear fruit. The white or cream-colored clusters of fruit, when they occur, are significant identifying characters, especially at the season after the leaves have fallen.

For convenience, these plants are discussed in this bulletin under three divisions: (1) Common poison-ivy, (2) oakleaf poison-ivy, and (3) Western poison-oak. A technical botanist would recognize many species of plants in what are included as the common poison-ivy, and anyone will recognize that various forms occur within this grouping.

COMMON POISON-IVY



Figure 1.—The shaded part shows the extensive area where some form of the common poison-ivy is likely to be found. Other forms also may occur in parts of the same region.

Some form of the common poison-ivy may be found in almost every part of the country but the extreme West. The shaded area on the map in [figure 1](#) shows the extensive range where some form of poison-ivy is likely to be found. The plant is known by various local names—poison-ivy, threeleaf ivy, poison creeper, climbing sumac, poison-oak, markweed, piery, and mercury. The most widespread form is classified by botanists under the botanical name of *Rhus radicans* or varieties of it; however, many of the variations are sometimes given other botanical names. The common poison-ivy may be considered as a vine in its most typical growth habit.



Figure 2.—Large poison-ivy vine growing on a tree for support. This is the usual form of the common poison-ivy in wooded areas.

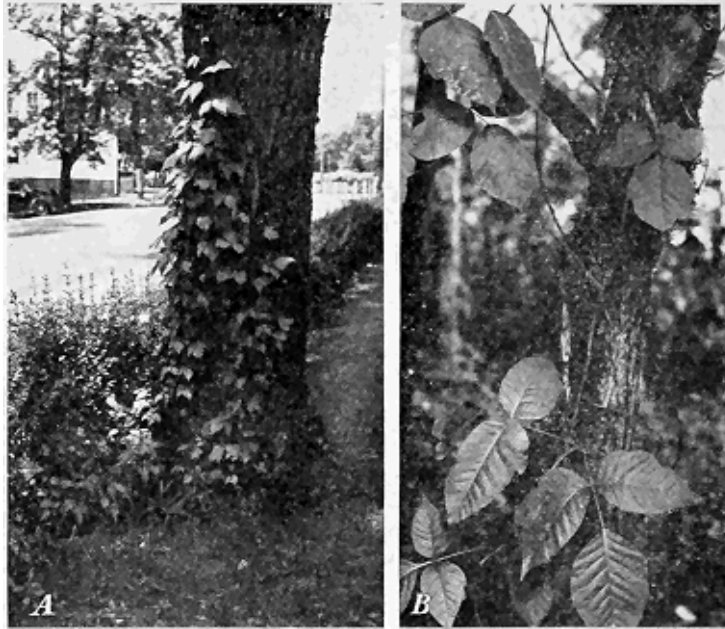


Figure 3.—A, Common poison-ivy growing in a hedge and on a shade tree on the edge of a lawn. B, Typical vine, showing leaves and the roots that attach it to the tree.

The vining type most often occurs in wooded areas ([fig. 2](#)), where it depends upon trees for support. Vines often grow for many years, becoming several inches in diameter and quite woody. Slender vines may run along the ground, mixed with shrubbery, or take support from a large or small tree. That a plant growing along the edge of a lawn and into the shrubbery may be quite inconspicuous as compared with a vine climbing on a lawn tree is shown in [figure 3](#). The vine readily develops roots when in contact with the ground or with any object that will support it. When the vines grow on trees these aerial roots attach the vine securely ([fig. 3. B](#)). A rank growth of these roots often causes the vines on trees to have the general appearance of a “fuzzy” rope, which sometimes serves a good purpose for identification. The vines and roots apparently do not cause injury to the tree except where growth may be sufficient to cover the supporting plant and exclude sunlight or break the plant from excessive weight. The vining nature of the plant makes it well adapted to climbing over stone walls or on brick and stone houses.

The fact that poison-ivy often becomes mixed in with ornamental shrubbery and vines often results in its cultivation as an ornamental vine by people who do not recognize the plant. An ivy plant growing on a house, as is shown in [figure 4](#), is often prized by an unsuspecting owner. The vine is attractive and quite effective as an ornamental, but such use should not be

tolerated, as it most certainly will result in cases of accidental poisoning and serve as propagating stock for more poison-ivy in the vicinity.



Figure 5.—Common poison-ivy growing along fence row. In open sunlight and over some large areas it is more of a shrub than a vine.

When poison-ivy becomes mixed in with other vines it is quite difficult to detect except by one who has carefully trained himself in observing and recognizing the plant. Also, some other vines and young plants resemble poison-ivy in many respects. The Virginia creeper and some forms of

the Boston ivy are often confused with it. The Virginia creeper can always be recognized by its five leaflets radiating from one point of attachment, as compared with the three leaflets of poison-ivy arranged in the same manner. The Boston ivy with three leaflets is sometimes difficult to detect; however, as a rule not all the leaves on the plant will be divided into three leaflets. By studying a large number of the leaves it is usually possible to detect on every Boston ivy plant some leaves that have only one deeply lobed blade or leaflet, while the poison-ivy always has the three leaflets. A number of other plants are easily confused with poison-ivy. No specific rule can be observed that will easily rule out all possibility of confusion. The best policy is to learn through practiced observation to know poison-ivy on sight and then to make the determination from observation of all parts of the plant, and never to judge from a single leaf or part.



Figure 4.—Common poison-ivy growing on side of house with ornamental shrubs. This accidental or intentional use of the attractive vine may lead to many cases of accidental poisoning.

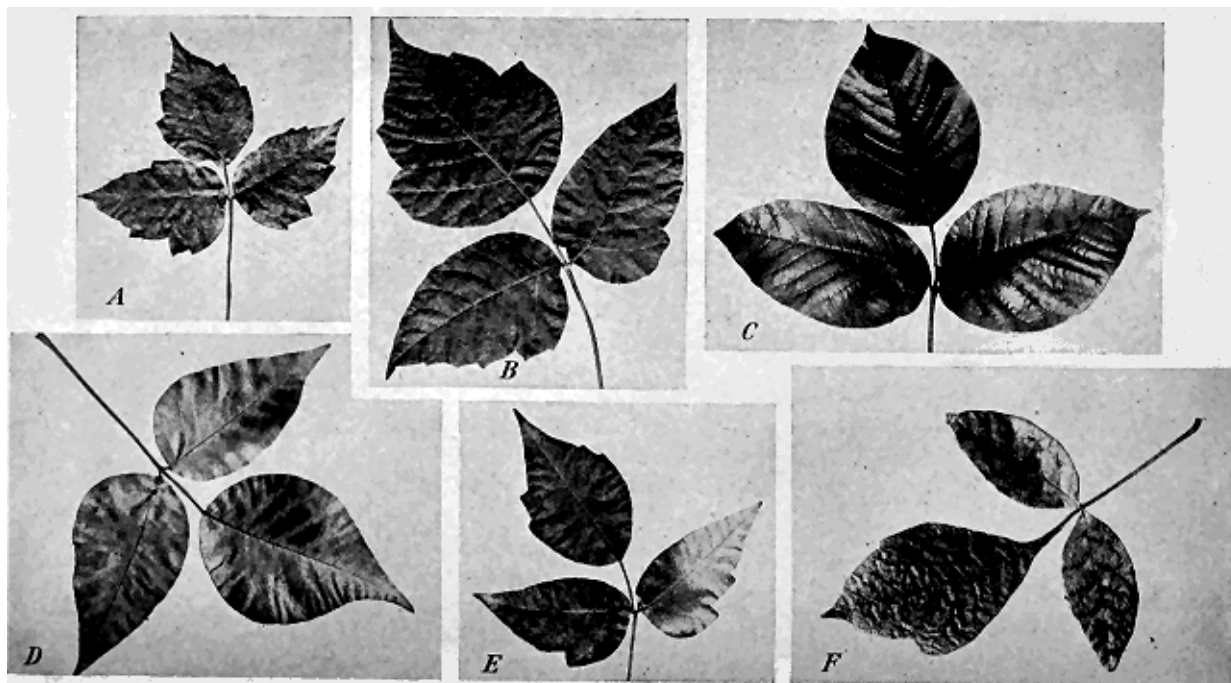


Figure 6.—Leaves of common poison-ivy, about half natural size. These may occur on the same or different plants: A, B, C and F, less common; D, with even margins, the most common and generally recognized form; E, with lobed margins, nearly nearly as often seen as the shape in D.



Figure 7.—Common poison-ivy vine with clusters of flowers in the axis of each leaf.

Common poison-ivy in full sunlight along fence rows or in open fields grows more in the form of a shrub than a vine. In some localities the common form is a low-growing shrub from 6 to 30 inches in height. This form, as well as the vining type, usually has a rather extensive horizontal system of rootstocks or stems at or just below the ground level. The vining form under some conditions later becomes a shrub. Plants of this type are shown in [figure 5](#), where the vine, supported on the fence, has extended upright stems that are

shrublike. In some localities the growth form over a wide range is consistently either a vine or a shrub type, while in others what seems to be the same species may produce either vines or shrubs.

The leaves of the common poison-ivy are extremely variable, the three leaflets being the only constant character. It is impossible to describe the great range of variation in the shape or lobing of the leaflets. A study of the leaf patterns, especially the six shown in [figure 6](#), will give a fair range of types, although other forms may be found. One plant may have a large variety of forms of leaves or it may have all leaves of about the same general character. The most common type of leaf having leaflets with even margins is shown in [figure 6, D](#), and one with leaflets having slightly lobed margins is shown in [figure 6, E](#). The other forms shown in [figure 6](#) are not quite so widespread, but may be the usual type throughout some areas.

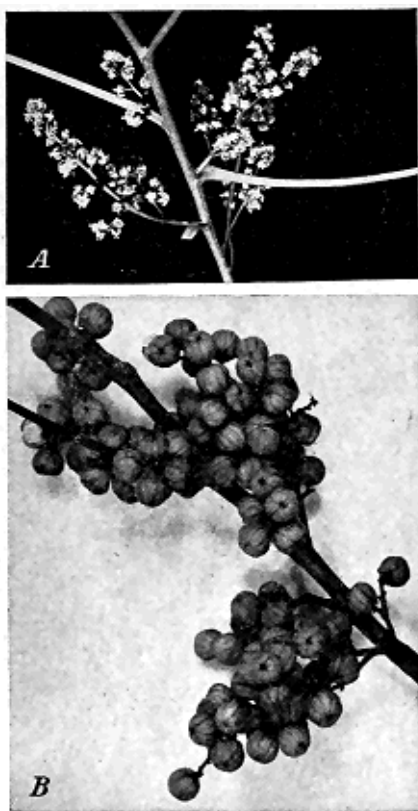


Figure 8.—Common poison-ivy, about natural size: A, Flowers; B, mature fruit.

Most of the vines or shrubs of poison-ivy produce some flowers that are always in clusters arising on the side of the stem immediately above a leaf. The flowers are rather inconspicuous among the leaves, as may be noted in [figure 7](#), but they are in quite distinct clusters, as shown in [figure 8, A](#). Frequently the flowers do not develop or are abortive and no fruit is produced. When fruits do develop they serve as a positive way of identifying the plant. The berries are not easily confused with the fruits of other plants. They are white and waxy in appearance and have rather distinct lines marking the outer surface, as the segments appear in a peeled orange. These lines may be noted in [figure 8, B](#), which shows dense clusters of mature fruit about natural size. Some forms of the poison-ivy have the fruit covered with fine hair, giving it a kind of downy appearance; however, in the more common form the fruits are entirely smooth. The fruit is especially helpful in identifying plants late in fall, in winter, and early in spring, when the leaves are not present.

OAKLEAF POISON-IVY

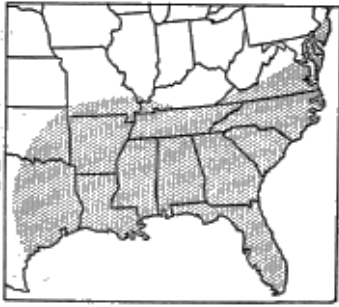


Figure 9.—The shaded area is the region where the eastern oakleaf form of poison-ivy is likely to occur. Other forms also may be in the same region.

Of the several kinds of poison-ivy, the oakleaf form occurring in the Eastern and Southern States is more distinctive than some other types. Some people know it as oakleaf ivy while others call it poison-oak, a name more commonly used for the western species discussed later. Botanists have not

always agreed upon the scientific name to be applied to this form; however, they now use the name *Rhus toxicodendron*. Some botanists have used the Latin names *R. quercifolia*, while others used the name *Toxicodendron quercifolium*, both of which are more descriptive of the oakleaf form, as the generic name of the oak is *Quercus*. The area where the oakleaf form may occur, from New Jersey to Texas, is outlined in the map shown in [figure 9](#). The common poison-ivy also occurs in some form throughout many parts of this same area.

The oakleaf poison-ivy usually does not climb as a vine, but occurs as a low-growing shrub. Stems are usually upright in their general growth habit, as they appear in [figure 10](#). The shrubs usually have rather slender branches, often covered with a fine pubescence that gives the plant a kind of downy appearance. The leaflets occur in threes, as in other ivy, but are lobed, somewhat on the general plan of the leaves of some kinds of oak. The middle leaflet usually is lobed somewhat alike on both margins and very much resembles a small oak leaf, while the two lateral leaflets are often irregularly lobed. The leaves shown in [figure 11](#) are a common form. The lighter color on the under side of one of the leaves ([fig. 11, A](#)) is due to the pubescence, or fine hairs, on the surface. The range in size of leaves varies considerably, even on the same plant; the leaf shown in [figure 11, B](#), however, is about natural size.



Figure 10.—Eastern oakleaf poison-ivy shrub. The upright growth as a low shrub is common. The three parts of each leaf somewhat resembles an oak leaf. The gall formations that cause wartlike protuberances are common on all species.

The fruit of oakleaf poison-ivy has the same general appearance as the fruit of common poison-ivy shown in [figure 8](#), although the individual fruits and stems are often pubescent, while most of the other forms have a waxy, smooth, cream-colored fruit.

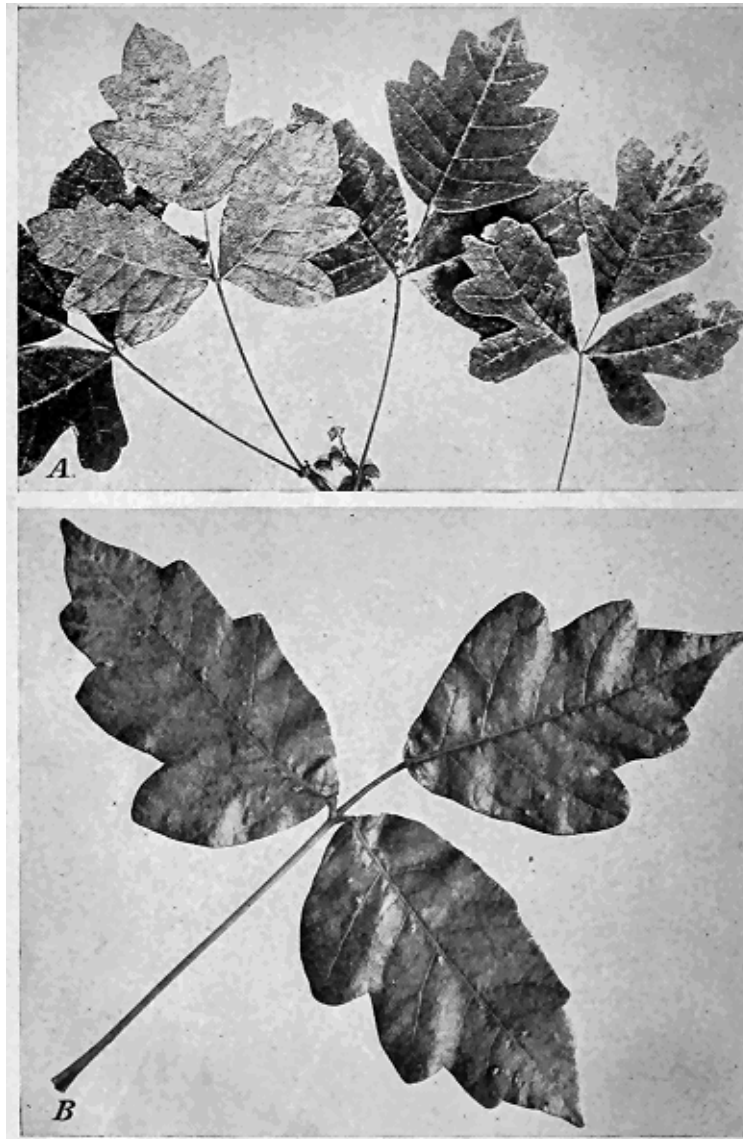


Figure 11.—Eastern oakleaf poison-ivy: *A*, Leaves, showing upper deep green surface and lower lighter colored surface. *B*, Single leaf, about natural size. The terminal leaflet more nearly resembles an oak leaf than the two lateral leaflets.

WESTERN POISON-OAK



Figure 12.—The shaded area is the region where the western poison-oak is likely to occur. Other forms may also be in the same region.

The western poison-oak (*Rhus diversiloba*) of the Pacific Coast States is usually known as poison-oak, but is occasionally referred to as poison-ivy or yeraa. It is likely to be found in the region designated by shading in [figure 12](#), and from central Oregon northward through Washington may be growing intermixed with common poison-ivy. The term “poison-oak” is a misnomer, as this species is in no way related to the oak but is related to poison-ivy. Anyone familiar with either of these plants in their several forms will likely recognize the other, both as plants to be avoided.

The most common growth habit of the western poison-oak is in the form of a rank upright shrub with many small woody stems rising from the ground. It frequently grows in great abundance along roadsides (see [fig. 16](#)) and in uncultivated fields or on abandoned land.



Figure 13.—Western poison-oak on a telephone pole. This vine form is not uncommon, especially on trees in woods.



Figure 14.—Western poison-oak: Upright shrub, with fruit, growing in open pasture field.

Sometimes western poison-oak attaches itself to upright objects for support and takes more or less the form of a vine. A plant of this type is shown in

[figure 13](#), where it is growing on a telephone pole by the roadside. It may be noted even in this case that the tendency is for individual branches to continue an upright growth rather than become entirely dependent upon other objects for support. In some areas, especially in woodland, poison-oak may grow principally as a vine. Areas have been noted where 70 to 80 percent of the trees leaves and flowers. in a wooded area support vines running more than 25 or 30 feet in height, although the most common form of the plant is the shrub.



Figure 15.—Western poison-oak, showing common leaf types: A, The more common leaves with irregular margins; B, less typical, although not uncommon, leaves with even margins.
Figure 15.—Western poison-oak, showing common leaf types: A, The more common leaves with irregular margins; B, less typical, although not uncommon, leaves with even margins.



Figure 16.—Western poison-oak growing alongside roadside. This rank shrubby growth is typical for most localities.

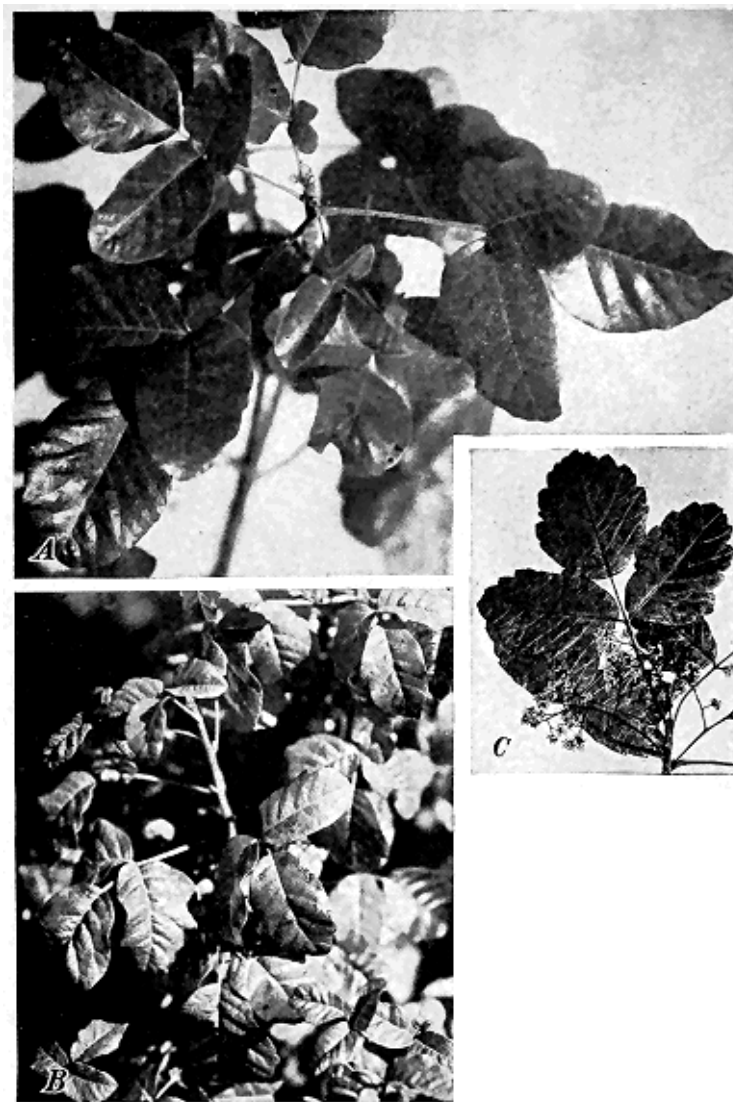


Figure 17.—Western poison-oak, showing common leaf types: *A*, Leaves with thick leathery appearance; *B*, leaves irregular in shape with much-curved or distorted surfaces; *C*, branch, with

In open pasture fields western poison-oak usually grows in spreading clumps from a few feet to several feet tall (figs. [14](#) and [16](#)). Where it develops extensively it greatly reduces the area for grazing and becomes a serious menace to most people who frequent the area or attend cattle that come in contact with the plants while grazing.



Figure 18.—Western poison-oak: *A*, Top of shrub, with leaves and fruit; *B*, clusters of fruit after leaves have fallen.

Low-growing plants, especially those exposed to full sunlight, are often quite woody and show no tendency for vining. Low-growing plants with woody stems are common in pasture areas or along roadsides. Livestock in grazing do not invade the poison-ivy shrub, and as a rule these plants continue to spread both by root-stock and seed.



Figure 19.—A large poison sumac shrub about 15 feet tall, growing on the edge of a swamp.



Figure 20.—The shaded area shows the region where poison sumac is likely to occur in bogs or swamps, although isolated plants are sometimes found in dry soil.

As in other ivy the leaves consist of three leaflets with much irregularity in the manner of lobing, especially of the two lateral leaflets. The Latin name *dirersiloba* refers to the irregularity occurring in the shapes of the leaves on different plants and on the same plant or even irregularity in the lobing of leaflets of the same leaf. Sometimes lobes occur on both sides of a leaflet, thus giving it somewhat the semblance of an oak leaf. The middle or terminal leaflet is more likely to be lobed on both sides and resembles an oak leaf more than the other two. A study of the leaves shown in figures [15](#) and [17](#) shows the variation in lobing. Some plants may have leaflets with an even margin and no lobing whatsoever, as those in [figure 15, B](#). The surface of the leaves is

usually glossy and uneven, so that they have a thick leathery appearance.

The flowers are borne in clusters on slender stems diverging from the axis of the leaf, as may be noted in [figure 17](#), C. The individual flowers are greenish white and about one-fourth inch across. The cluster of flowers matures into greenish or creamy-white berrylike fruits about mid-October. These are about the size of small currants and are much like other poison-ivy fruits, having a smooth greenish-white glossy surface striped somewhat like the segments of a peeled orange. Many plants bear no fruit, although others produce it in abundance, as shown in [figure 18, A](#). The fruits are not always spherical, but sometimes have a somewhat flattened appearance, as is shown in [figure 18, B](#). They remain on the plants throughout fall and winter and are quite helpful in identifying poison-oak in seasons after the leaves have fallen.

POISON SUMAC

Poison sumac grows as a coarse woody shrub or small tree ([fig. 19](#)) and never in the vinelike form of its poison-ivy relatives. This plant is known also as swamp sumac, poison elder, poison ash, poison dogwood, and thunderwood. It does not have variable forms, such as occur in poison-oak or poison-ivy, and botanists agree to call it *Rhus vernix*. The area in the Eastern States where it is likely to be found is chiefly eastward from eastern Minnesota, northeastern Illinois, Indiana, central Kentucky and Tennessee, and southeastern Texas, as shown on the shaded area of the map, [figure 20](#). This shrub is usually associated with swamps and bogs, and the most typical growth occurs along the margin of an area of wet acid soil.



Figure 21.—Small branch of poison sumac with six compound leaves.

The plants range in height from 5 or 6 feet to small trees that may attain a height of 25 feet. The poison sumac shrub shown in [figure 19](#) is growing on the edge of a swamp and is quite typical of the general appearance of the larger shrubs, which approach a treelike form. As a rule the shrubs do not have a symmetrical upright treelike appearance. They are more or less inclined to lean and have branched stems with about the same diameter from the ground level to the middle height of the shrub.

Occasional isolated plants are found outside of swampy regions. Apparently these plants are started from seed distributed by birds. The plants in dry soil usually do not become more than a few feet tall. They are likely to

cause poisoning to unsuspecting individuals, as single isolated plants are not readily recognized, especially out of their usual region.

The leaves of the poison sumac are divided into 7 to 13 leaflets, arranged in pairs with a single leaflet at the end of the midrib ([fig. 21](#)).

The leaflets are an elongated oval shape without teeth or serrations on the margins. They are 3 to 4 inches long and 1 to 2 inches wide, with a smooth velvetlike texture and bright orange color when they first appear in spring. Later they become dark green and glossy on the upper surface and pale green on the lower and have scarlet midribs. Early in fall they turn to a brilliant red-orange or russet shade.

The small yellowish-green flowers are borne in clusters in slender stems arising from the axis of leaves along the smaller branches. The slender clusters of flowers have much the general appearance of the poison-ivy flower, but hang in much longer clusters. The flowers mature into ivory-white or green-colored fruits resembling those of poison-oak or poison-ivy, except that they are usually less compact and hang in loose clusters that may be 10 to 12 inches in length ([fig. 22](#)).

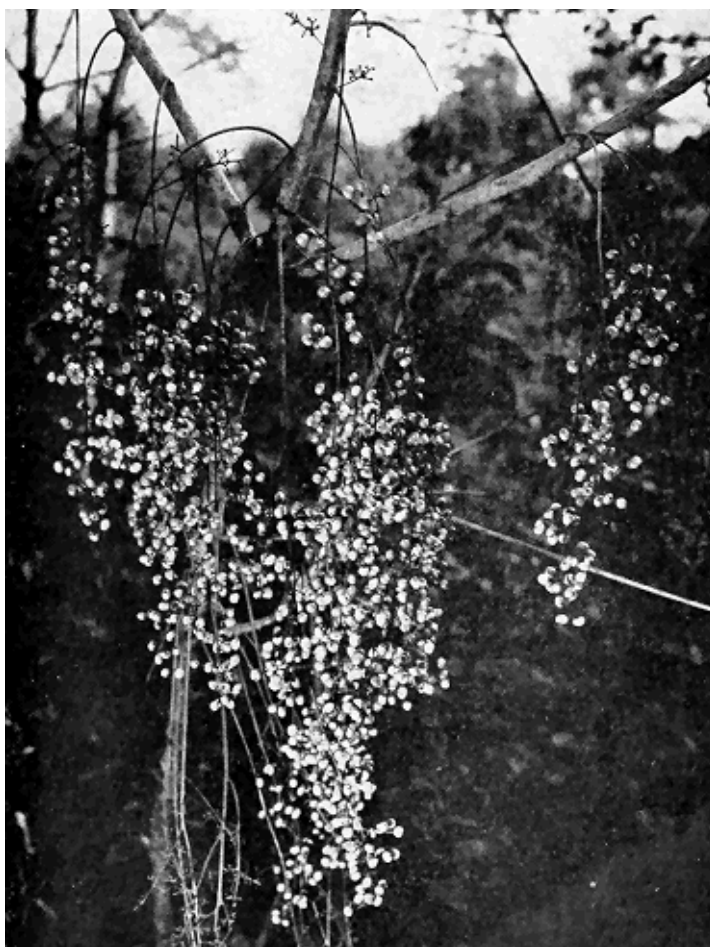


Figure 22.—Fruit of poison sumac, one-third natural size.

These fruits somewhat resemble those of poison-ivy. They are always on drooping slender stems attached at the side of the small branches and are never terminal as in the species of sumac that are not poisonous.

CONFUSION OF POISONOUS AND HARMLESS SUMACS

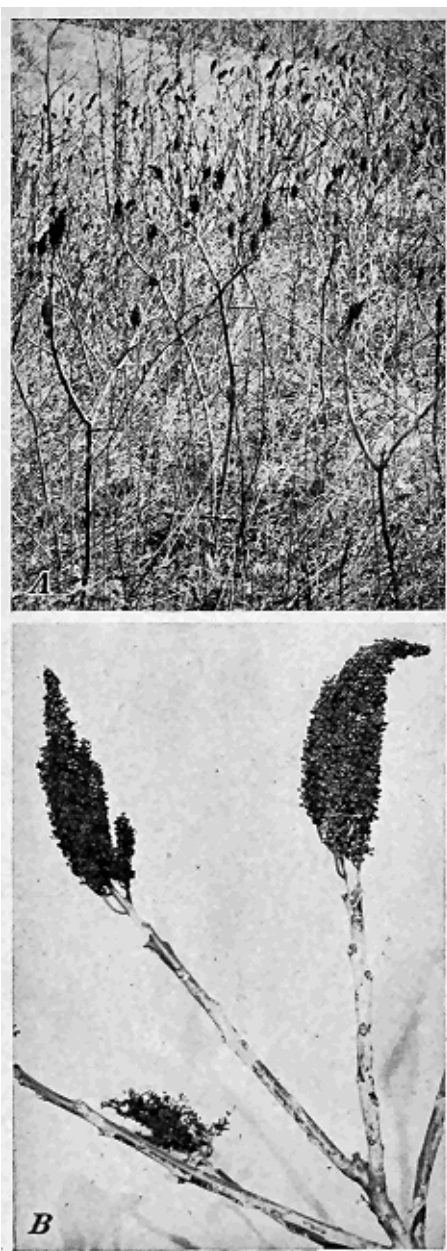


Figure 23.—A, Smooth sumac and dwarf sumac growing in a mixed stand. These plants are not poisonous. B, Terminal fruiting spike of smooth sumac, about one-seventh natural size. This form of terminal fruiting spike is typical of all the species of sumac that are not poisonous.

Because of the same general appearance of several common species of sumac and the poison sumac, considerable confusion has occurred as to which one is poisonous. Throughout most of the range where poison sumac grows, three other species are the only ones that are likely to be confused. These are the smooth sumac (*Rhus glabra*), staghorn sumac (*R. typhina*), and dwarf sumac (*R. copallina*). All the nonpoisonous species have red fruits that together form a distinctive terminal seed head, as shown in [figure 23](#). These are easily distinguished from the slender hanging clusters of white fruit of the poison sumac, as shown in [figure 22](#). Sometimes more than one species of the harmless sumac grow together, as shown in [figure 23, A](#).

When seed heads or flower heads occur on the plants it is easy to distinguish the poisonous from the harmless plants; however, in many clumps of either kind, flowers or fruit may not develop. Fortunately, the leaves have some rather distinct characteristics, which can be easily observed. [Figure 24](#) shows leaves of the three harmless species as compared with poison sumac.

The leaves of the smooth sumac ([fig. 24, A](#)) and of the staghorn sumac ([fig. 24, B](#)) have many leaflets, which are slender lance-shaped with a toothed margin. In these species there are usually more than 13 leaflets. The leaves of the dwarf sumac ([fig. 24, C](#)) and the poison sumac ([fig. 24, D](#)) have fewer leaflets, and these are more oval-shaped, with smooth or even margins.

The dwarf sumac, however, may be readily distinguished from the poisonous sumac by the winged midrib of the leaf ([fig. 24, E](#)). There is

considerable variation in the size of the wing margin along the midrib, which in some cases may be reduced almost to a line between the leaflets. The winged midrib is more prominent near the terminal leaflet and can always be detected. The midrib of the poison sumac is never winged. A little study of the fruit and the leaf characteristics of the poisonous and harmless species will make it possible to avoid the poisonous one and utilize the other, which has considerable value as an ornamental plant, as a source of commercial tannin, and for controlling erosion on waste hillsides.

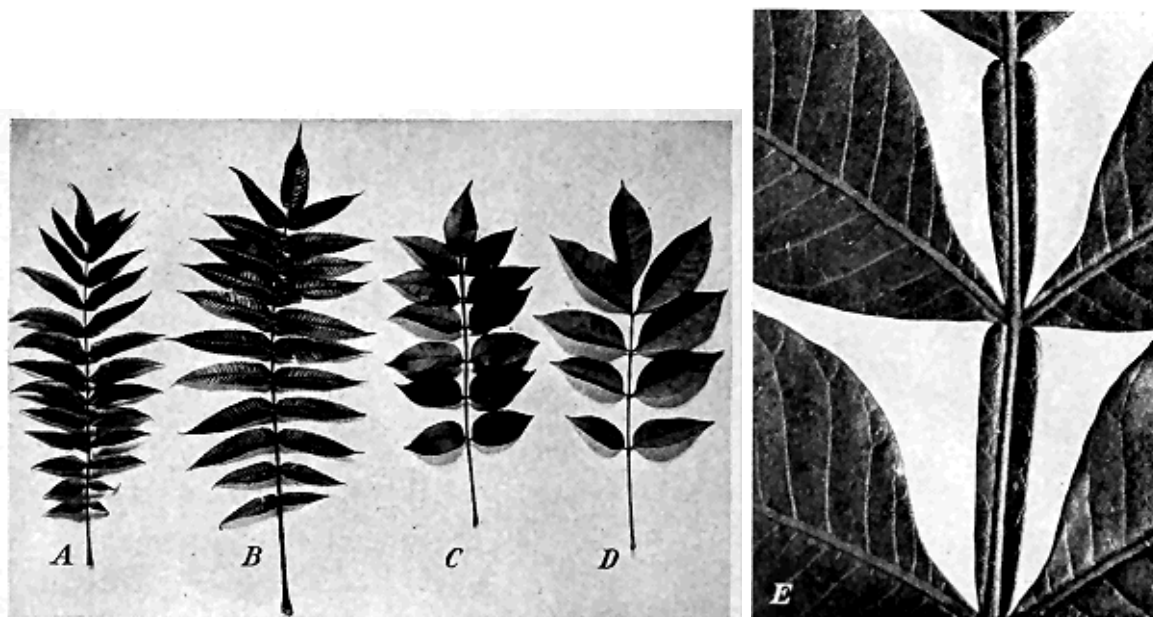


Figure 24.—Leaves of species of sumac that are often confused: A, Smooth sumac; B, staghorn sumac; C, dwarf sumac; D, poison sumac; E, enlarged portion of dwarf sumac leaf from C, showing the wing margin of the midrib. Poison sumac does not have the winged midrib.

INTRODUCED POISONOUS SUMAC AND RELATED SPECIES

The small Japanese lacquer-tree, although uncommon in the United States, is an introduced species of sumac (*Rhus verniciflua*) that is quite similar to the native poison sumac. It is native to Japan and China and is the source of Japanese black lacquer. Cases of poisoning have been traced to contact with lacquered articles. Because of its poisonous properties, this tree should never be planted.

A native shrub or small tree called poison-wood, doctor gum, or coral sumac is commonly found in the pinelands and hummocks of extreme southern Florida, the Keys, and the West Indies. It is much like the poison sumac and closely related to it. Most botanists name it *Metopium toxiferum*, although previously it was named *Rhus metopium*, and sometimes metopium is used as its common name. The small tree, or shrub, has the same general appearance as poison sumac. The leaves have the same general appearance, except that there are usually only three to seven leaflets, which are more rounded. The fruits are borne in clusters in the same manner as those of poison sumac ([fig. 22](#)), but individual fruits are two or three times as large and orange-colored. All parts of the plant are exceedingly poisonous to the touch and cause the same kind of skin irritation as poison-ivy or poison sumac.

POISONING

Through experience, many people know that they are susceptible to poisoning by poison-ivy, poison-oak, or poison sumac, while some others either have escaped contamination or have a certain degree of immunity. The extent of immunity appears only relative, and absolute immunity to the toxic principle apparently does not exist. Persons who have shown a degree of immunity upon repeated contact with the plants may develop poisoning on subsequent exposure.

The skin irritant of poison-ivy, poison-oak, and poison sumac is the same toxic agent. It is a nonvolatile phenolic substance called urushiol and is found in all parts of the plant, including roots and fruit. It occurs in great abundance in the plant sap. The danger of poisoning is greatest in spring and summer, when the sap is abundant, and least late in fall or in winter.

Poisoning is usually caused by contact with some part of the plant. A very small quantity of the poisonous substance is capable of producing severe inflammation of the skin and can easily be transferred from one object to another. Clothing may become contaminated and is often a source of such prolonged infection that it is likely to be judged as a case of poisoning difficult to cure. Dogs and cats frequently touch the plants and transmit the poison to unsuspecting persons. The poison may remain on the fur of animals for a considerable period after they have walked or run through poison-ivy plants. Smoke from burning plants will carry the toxin and has been reported to cause severe cases of poisoning. Cases of poisoning of children from eating the fruit have been reported. A local belief that eating a few leaves of these plants will develop immunity in the individual is unfounded. It never should be attempted. No part of the plant should ever be taken internally, as it is a violent irritant and poisonous to man.

Cattle, horses, sheep, hogs, and other livestock apparently do not suffer from skin irritation caused by these plants, although they have been observed to graze upon the foliage occasionally. Bees collect honey from the flowers, though no reports are on record of any ill effects from the use of the honey.

The time between contamination of the skin and the first symptoms varies greatly with individuals and probably with conditions. The first symptoms of itching or burning sensation may develop in a few hours or after 5 days or even more. The delay in development of symptoms is often confusing in attempting to determine the time or location when contamination occurred. The itching sensation and subsequent inflammation, which usually develops into water blisters under the skin, may continue for several days from a single contamination. Persistence of symptoms over a long period is likely to be due to new contacts with plants or with previously contaminated clothing or animals. Severe infection may produce more serious symptoms, which result in much pain through abscesses, enlarged glands, fever, or complicated constitutional malfunction. Secondary infections are always a possibility in any break in the skin, such as is produced by breaking vesicles that have formed as large water blisters.

PRECAUTIONS AGAINST POISONING

The most effective way to prevent poisoning from the plants is to avoid contact with them. If it is necessary to work among them, some measure of prevention can be gained by wearing protective clothing, but it is necessary to remember that the active poisonous principle can easily be transferred. Some protection also may be obtained in advance by the use of protective creams or lotions. They prevent the poison from touching the skin, or make the active principle easily removed, or neutralize it to a certain degree.

Various formulas containing ferric chloride and glycerin have been used more or less effectively as a preventive measure, although there may be some danger of pigmentation of the skin from the ferric chloride. A common formula is ferric chloride, 5 parts; glycerin, 25 parts; alcohol, 25 parts; water. 50 parts.

A lotion of this formula applied to the exposed skin prior to contact with poisonous plants gives effective protection for many people. It is especially effective if followed by thorough washing with soap and water. Such washing, even if no protective lotion has been used, is often effective in preventing poisoning. Proper precaution should be taken by repeated lathering with a strong alkali soap, followed by thorough rinsing and repeating the process several times. The water should be frequently changed, and a shower or flowing water bath is preferable. The soap probably only emulsifies the active

principle, and thorough rinsing is necessary to avoid spreading the poison to other parts of the body.

The United States Public Health Service reports successful chemical protection against ivy poisoning by use of an oxidizing agent—sodium perborate prepared and used fresh as an ointment. Following is a formula given for the ointment: Cetyl alcohol, 35.1 percent; stearyl alcohol, 5.3; ceresin, 3.5; castor oil, 20.8; mineral oil, 21.9; Duponol WA pure, 1.7; sodium perborate, 10.0; and boric acid, 1.7 percent.

A more recent less oily formula recommended for both mechanical and chemical protection is given as follows: Shellac, 13 parts; isopropanol, 31; linseed oil, 4; titanium oxide, 12; sodium perborate, 13; talcum, 20; and carbitol, 3 parts.

Sometimes cottonseed oil, olive oil, or petrolatum is applied as a protective ointment when contamination is anticipated. Later the oil must be completely removed by repeated washing.

Clothing, wearing apparel, and tools that have been contaminated are often difficult to handle without further danger of poisoning. Some who are more or less immune to poisoning often contaminate others by carelessness. Contamination on automobile door handles or on a steering wheel after a trip to the woods often causes prolonged cases of poisoning of persons who have not been near the plants. One of the most effective ways for decontaminating articles is by thorough washing through several changes of strong soap and water. Contaminated clothing should not be worn again until thoroughly washed. It should not be washed with other clothes, and care should be taken to rinse thoroughly any implements used in washing.

Dogs and cats can be decontaminated in the same manner; precaution should be taken, however, to avoid being poisoned in doing the washing. It is likely that most “dry cleaning” processes will remove any contaminant; but there is always danger that clothing sent to commercial cleaners may cause poisoning to unsuspecting employees.

Certain prophylactic inoculations for the prevention of ivy poisoning have been developed and used with limited success; such treatment, however, should be administered by a physician only. The theory that eating a few leaves of these poisonous plants will confer immunity is without foundation; it should

not be attempted even as an experiment, because very serious poisoning is likely to result.

TREATMENT FOR POISONING^[1]

^[1] For a fuller description of treatments for poisoning see: United States Public Health Service, ivy and sumac poisoning. Pub. Health Rpts. Sup. 161 (rev.), 8 pp., 1943. For sale by the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price, 10 cents.

There seems to be no absolute quick cure for ivy poisoning for all individuals, even though many studies have been made to devise effective remedies. Remedies may be helpful in removing the poisonous principle or rendering it inactive and for giving some relief from the irritation. Mild poisoning usually subsides within a few days, but if the inflammation is severe or extensive a physician should be consulted. Self-treatment also has other dangers, as the symptoms of ivy poisoning may be confused with those of other conditions and harm may be done by improper treatment. In all cases the safest procedure is to consult a physician.

Some tried remedies are effective on certain individuals and ineffective on others. Sometimes a person will find one remedy effective and later, to his surprise, get very little relief from it. A large number of patent medicine remedies of doubtful value are commonly offered for sale. The following are more or less standard measures that have been advocated by different individuals and found helpful in certain cases.

The United States Public Health Service recommends a 10 percent alcoholic solution of tannic acid applied as a lotion. It is stated that in using it the tops of the blisters should be rubbed off with sterile gauze saturated with the solution and that any large blisters should be opened with a sterile instrument. The treatment should be repeated three or four times at 6-hour intervals. A modified form of this treatment recommends vigorous rubbing of the lesions with alcohol-soaked gauze to remove the tops of blisters, followed by application of a 10 percent aqueous solution of tannic acid as a lotion. Any treatment involving opening of blisters and the application of alcohol is severe, and many cases requiring such treatment should be handled by physicians.

One of the oldest well-established remedies is a fresh solution of one 5-grain tablet of potassium permanganate dissolved in 1 quart of water;

concentrations as much as 5 percent, however, have been used. The potassium permanganate is apparently active in neutralizing any poison that may remain on the surface of the skin and it has some therapeutic properties. This solution may be applied freely to the irritated or contaminated skin, but the skin should be rinsed with water a few minutes after the stronger solutions are used. Applications should be repeated every hour or two. The brown stain it causes on the skin will in time wear off, or it may be removed with lemon juice. It more or less permanently stains clothing.

Ferric chloride in combination with several other substances has been used effectively by many people. One of the most common formulas is that given under the discussion of precautions against poisoning ([p. 22](#)). To this formula 1 to 3 percent of phenol is frequently added to reduce the itching; however, not more than 1 percent of phenol should ever be used without the advice of a physician, on account of the danger of systemic absorption and poisoning. The solution should be freely applied to the irritated or contaminated skin every hour or two. Some instances of persistence of the brown discoloration of the skin have been reported. For this reason its use is limited, but for most individuals the discoloration completely disappears after a few days. Stain on clothing is usually permanent.

Photographer's hypo solution has been used with good results in many cases. This is a 10-percent solution of sodium thiosulfate in water. The irritated or contaminated skin should be bathed in it every hour or two. It does not stain skin or clothing.

A calamine lotion with the addition of about 2 percent of phenol is often used and in many cases is effective. Frequent applications are necessary. For some people it is effective in relieving the itching. The pink stain is not permanent.

It is better to leave inflamed areas exposed to the air rather than cover them with dressings. If poisoning is so severe that dressings appear necessary, a physician should be consulted before any home remedies are applied.

ERADICATION OF PLANTS

Poison-ivy^[2] can be killed either by mechanical means, such as digging, or by applying chemical herbicides. The best modern herbicides are so deadly to plants that a single treatment kills all the leaves and 90 percent or more of the roots. This is the easiest and least dangerous way to destroy these poisonous pests. There are places, however, where chemicals cannot be used, as in hedges and shrubbery where the poison-ivy is closely mixed with the valuable plants. In such situations, hand-pulling is the only satisfactory method. At times also chemicals and spraying equipment are not available, and then the mattock, scythe, or plow must be resorted to. Even with these simple tools some timing and handling methods are better than others.

^[2] For brevity, the single designation poison-ivy is used under this heading to apply equally to all poison-ivies and to poison sumac.

CHEMICAL WEED KILLERS

By using care poison-ivy can be destroyed with chemicals without danger of poisoning to the operator. Except in very heavy growth, one may stand at a distance from the plants and apply the herbicide without touching the leaves. Most chemicals are applied as a spray solution, and many sprayers can be equipped with extension nozzles 2 feet or more in length. The greatest danger of ivy poisoning occurs from careless handling of gloves, shoes, and clothing after finishing the work. Garments should be worn that can either be destroyed after use or washed thoroughly in hot soda solution and soapsuds.

The chemicals thus far found most satisfactory on poison-ivy are, in approximate order of effectiveness, ammonium sulfamate, 2,4-D (2,4-dichlorophenoxyacetic acid), ammonium thiocyanate, borax, carbon disulfide, coal-tar creosote oil, fuel oil and similar petroleum distillates, sodium chlorate, and sodium arsenite. Less effective chemicals include common salt, kerosene, gasoline, calcium cyanamide, ammonium sulfate, and iron sulfate.

Chemicals other than borax are best applied as a spray to the foliage. Borax is scattered dry on the soil around the plants. Any field or garden spray

machine or even a sprinkling can is useful for applying the liquids, but a common knapsack sprayer holding about 2 gallons is convenient and does not waste the solution, application should be as a fine spray under moderate pressure rather than a driving mist, since the object is to wet the leaves of the poison-ivy and avoid wetting the leaves of desirable plants.

Most chemicals used on poison-ivy are injurious to the foliage of all kinds of plants and must be applied with caution if the surrounding vegetation is valuable. During the early part of the growing season the leaves of the poisonous plants usually tend to stand conspicuously apart from those of adjacent plants and can be treated separately if sprayed with care. Later the leaves become intermingled and injury to adjacent species is unavoidable. Chemicals other than oil are not injurious to the thick bark of an old tree, and poison-ivy clinging to the trunk can be safely sprayed with them.

Chemical sprays can be used at any time when poison-ivy is in full leaf, June and July being perhaps the best months. Ordinarily treatments should begin not later than August 15, as poison-ivy then begins to go dormant and sprays are ineffective.

Best results are obtained both with sprays and with dry chemicals when the soil is moist but not wet at time of treatment. Borax can well be applied just before a rain, but sprayed chemicals are likely to be washed off if rain falls within 24 hours. One gallon of spray solution is sufficient to cover all the leaves on 1 square rod (272 square feet) of dense poison-ivy, or 2 to 4 square rods of a scattered stand.

Sprays give best results when applied in early morning or late afternoon when the air is cool and moist. Spraying in the middle of a dry, hot day is likely to be disappointing unless the plants are in deep shade.

No method of chemical eradication can be depended upon to kill all the plants in a stand of poison-ivy with one application. Three to four light retreatments made as soon as the new leaves are fully expanded are always necessary (1) to destroy plants missed the first time, (2) to treat new growth from the old roots, and (3) to destroy seedlings. At least three and sometimes four treatments at intervals of 2 to 8 weeks are necessary before all plants are dead. These followup treatments do not require much time or material, but neglect of them may easily lead to serious reinfestation.

Poison-ivy has an annoying habit of “playing possum.” Plants believed dead sometimes revive after many months. Thus an area under treatment must be watched closely for at least a year if complete eradication is to be assured.

Dead foliage and steins remaining after the plants have been killed with chemicals are only slightly poisonous, but to avoid difficulty the dead stems should be cut off and burned.

Ammonium sulfamate.—This chemical, under a proprietary name, has come on the market recently as a weed killer and has proved especially useful against poison-ivy. Ammonium sulfamate is not dangerous to use under ordinary conditions, is not harmful to the soil, and is not flammable. It is slowly corrosive to metals, and spray equipment should be washed thoroughly with a solution of lime or soda immediately after use. Like salt, ammonium sulfamate absorbs moisture from the air and must be stored in a dry place.

Ammonium sulfamate is dissolved in water at the rate of 12 ounces per gallon and sprayed on the leaves until they are wet but not dripping. The foliage begins to wilt within 21 hours, but the action is slow, and a week or more may elapse before the leaves die. Some new growth may be expected after 4 to 8 weeks, and this should be sprayed as soon as the new leaves are fully expanded. Two or occasionally three such retreatments are needed before the kill is complete.

2,4-D (2,4-dichlorophenoxyacetic acid).—This chemical under the abbreviated name 2,4-D, or DCP, is a recent addition to the list of effective weed killers and may become an important weapon for use against poison-ivy. It should not be confused with DDT or DD, two other new agricultural chemicals, the first an insecticide and the second used as a soil fumigant. Under wartime restrictions the use of 2,4-D was very limited, but its prospective value is so great that those interested in the control of poison-ivy should be informed about it.

The chemical 2,4-D is one of the so-called “plant hormones,” or “growth regulators.” In minute quantities it stimulates some of the plant functions, but in slightly larger dosage it is toxic. A solution consisting of one-seventh ounce in 1 gallon of water sprayed on poison-ivy foliage kills all leaves hit and may kill the roots. In the concentration used in weed killing 2,4-D is not believed to be harmful to animal life, but until this point can be thoroughly clarified the chemical should be used with reasonable caution to avoid poisoning. 2,4-D is

not flammable and is not corrosive to spray machinery. If the sprayer is used also for insecticides and fungicides it must be washed very thoroughly to remove the slightest trace of the 2,4-D. This is very difficult, however; it is safer to use another sprayer.

To make a spray solution 2,4-D must be combined with other chemicals, as it is not soluble in water. A number of substances, including polyethelene glycols and naphthenic soaps, have been used for this purpose, but the best solvent has not yet been determined. Commercial weed killers containing 2,4-D are of varying composition.

Ammonium thiocyanate.—This byproduct of the coal-gas industry is one of the newer herbicides. The crude commercial chemical is a pink or reddish salt, very soluble in water. It does not keep well, tends to disintegrate into a red, sludgy mass, and is difficult and disagreeable to handle. A purer form in clean white crystals is available and is well worth the slight extra cost. In either form ammonium thiocyanate is injurious to the skin and to leather articles, and in time it corrodes metals. The spray solution is made by dissolving 1 to 1½ pounds of ammonium thiocyanate in 1 gallon of water.

Most annual, plants and some perennials are killed by a single application of ammonium thiocyanate. Poison-ivy usually requires three or four treatments. Ammonium thiocyanate has a high content of nitrogen, much of which remains in the soil and eventually becomes a fertilizer. Thus an area treated with this chemical is likely to produce a profuse growth of annual weeds a month or two after treatment.

Borax.—Common borax scattered on the soil is deadly to the roots of many plants. Spread dry at the rate of 10 to 15 pounds per square rod, it kills poison-ivy slowly over a period of 6 to 8 weeks. A second somewhat lighter treatment 3 to 5 months later is usually necessary to kill all plants completely.

Borax is easy to handle, nonpoisonous, and nonflammable, readily obtainable, and not expensive. It is one of the best chemicals to use around stone walls, rocky ledges, fences, and other inaccessible places.

Plants differ greatly in their resistance to borax. Some species, as Virginia creeper and deep-rooted trees, do not seem to be injured, but others, notably dogwood, are very sensitive to it. Until more complete information is available as to the effect of borax on other plants, it should be employed on poison-ivy only where possible damage to surrounding vegetation is not objectionable.

Carbon disulfide.—The fumes of this pungent, volatile liquid are very poisonous to plants, and the chemical is widely used in the West to kill deep-rooted perennial weeds. When poison-oak, particularly, is growing in a mixture with wild morning-glory (bind-weed) or similar noxious weeds, both plants may often be killed at one operation by treating the soil with carbon disulfide. Holes ordinarily 6 inches deep and 18 inches apart in all directions are punched in the soil with a 1-inch iron bar, and 2 ounces of carbon disulfide are poured in each hole. The openings are closed immediately with a wooden club. This dosage equals 20 pounds of the chemical per square rod. Carbon disulfide is likely to be less effective on shallow-rooted than on deep-rooted plants and is therefore of limited value against poison-ivy, which is surface-rooted.

Carbon disulfide is poisonous, and as it is also highly flammable it should be handled as carefully as gasoline.

Coal-tar creosote oil.—In the fractional distillation of coal, many oils and oillike substances are obtained, some of which are very toxic to plants. The most common and usually the cheapest is the creosote used for wood preservation. This substance has great penetrating power and is an excellent herbicide on woody plants like poison-ivy. Ordinarily creosote oil is too expensive to use alone and is therefore mixed with some kind of petroleum oil. A mixture of 75 percent fuel oil and 25 percent creosote oil is almost as effective as straight creosote and is made at one-third the cost. Two applications of this mixture at an interval of about 6 weeks usually kill a high percentage of poison-ivy. Creosote is not selective and kills all plants alike.

Fuel oil (Diesel and similar oils).—Crude petroleum is refined into many kinds of oil, and those corresponding to fuel oil. Diesel oil, stovetop oil, and orchard-heating oil are more injurious to plants than crude oil itself or the higher distillates, such as gasoline and kerosene. Fuel oils are widely used for destroying vegetation, particularly by highway departments in States having oil resources. When sprayed on poison-ivy, fuel oil brings about a slow but fairly sure death of the plants. Two or three applications may be necessary before the roots are killed. Fuel oil, like creosote, must be used with care around valuable trees and shrubs because it penetrates the bark and kills or injures all species. As used for poison-ivy spraying there is no injury to the soil. The oil-soaked vegetation is flammable.

Sodium chlorate.—Sodium chlorate is the standard herbicide for killing deep-rooted perennial weeds like bindweed, but when used as a spray, it is a

dangerous fire hazard, and when applied dry, it sterilizes the soil for 2 to 4 years. Where sterilization of the soil is desirable or not objectionable, one application of the dry chemical, evenly at the rate of 3 to 4 pounds per square rod, is sufficient for a complete kill.

If used as a foliage spray on poison-ivy, the chemical is dissolved in water at the rate of 8 to 12 ounces per gallon. The leaves are killed within a few hours, but the plants are likely to make a strong new growth and may require as many as six treatments before they cease to produce new leaves.

Sodium chlorate mixed with wood, cloth, or other organic materials is highly combustible and easily ignited. Shoes and clothing on which chlorate solution has been allowed to dry are especially dangerous, for they will ignite and burn with explosive fury. Contaminated clothing should be kept wet until thoroughly washed in a large volume of water. Chlorate should be stored only in original metal containers, as any admixture of chlorate, wet or dry, with straw, wood, dust, cloth, or leather has properties similar to those of gunpowder or matches.

Sodium arsenite.—Sodium arsenite is one of the most powerful plant poisons known and is widely used for sterilizing soil on railroads, roadways, and other places where no vegetation of any kind is desired. It is relatively inexpensive. In normal times sodium arsenite is obtainable in both powder and liquid form. In weak solution (4 to 6 ounces per gallon of water) the chemical can be used as a spray to destroy poison-ivy. Like sodium chlorate spray, it kills the leaves but not the roots, and five or six treatments may be required. The solution kills the leaves of all plants impartially and will also kill the young tender bark of shrubs and trees. This may be either an advantage or a disadvantage, depending upon circumstances.

All compounds containing arsenic are deadly poisons if taken internally. Arsenical sprays and chemicals should not be used where there is any possibility that materials sprayed with them will be eaten by animals or man. In some communities special permission must be obtained before arsenical weed killers may be used.

ERADICATION BY MECHANICAL MEANS

Poison-ivy can be grubbed out by hand quite readily early in spring and late in fall. When the ground is soft after rains the roots come out in long pieces. Grubbing when the soil is dry and hard is almost futile, since the roots break off in the ground, leaving large pieces that later sprout vigorously. Eradication by grubbing is permanent if well done. Because of his close contact with the plants, the person doing the grubbing should have a high degree of immunity to ivy-poisoning. Many people are not so immune as they believe, and it is common sense to wear leather gloves with gauntlets and a shirt with long sleeves. If care is taken to prevent the poison-ivy from touching the face, and if the clothes, including the gloves, are burned or thoroughly washed after use, severe poisoning may be avoided.

Poison-ivy vines climbing on trees should be severed at the base and as much of the vine as possible pulled away from the tree. Often the roots of the tree and weed are so intertwined that grubbing is impossible without injury to the tree. Close mowing of the poison-ivy shoots at frequent intervals is the only remedy. Roots and stems removed in grubbing should be burned or otherwise destroyed, since the dry material is almost as poisonous as the fresh. Care must be taken to keep out of the smoke.

Old plants of poison-ivy produce an abundance of seeds, and these are freely disseminated, especially by birds. A poison-ivy seedling 2 months old usually has a root that one mowing will not kill. Seedling plants at the end of the first year have well-established underground runners that only grubbing or chemicals will kill. Seedlings are a constant threat as long as old poison-ivy is in the neighborhood.

Plowing is of little value in combating poison-ivy unless followed by persistent stirring with a cultivator or harrow, to keep all roots loosened from the soil. Otherwise plowing merely propagates the weed.

Mowing with a scythe or sickle is a common method of attacking poison-ivy, but it has little effect on the roots unless frequently repeated. The number of cuttings required to kill depends upon the age of the plants and the size of the roots, but is seldom less than four. New shoots should be cut as soon as the first leaf on each stem is full size. Cutting either earlier or later delays eradication.

Weed burners are implements resembling oversize blowtorches. Two types are manufactured, one producing a short, hot, blue flame and the other a long

smoky-yellow flame. Both are useful for destroying foliage. Neither is likely to kill perennial plants like poison-ivy at one operation, even though the flame is held on the weeds for a minute or more until leaves and stems are completely incinerated. The roots are not injured by this, and new leafy shoots soon develop. A better plan is to “sear” the plants by passing the flame slowly over the vines just long enough to sear or wilt the leaves, but not long enough to burn them. The roots appear to be injured more by “searing” than completely burning the leaves, and the operation is quicker and cheaper. Three to four searings when the leaves reach full size usually kill the roots.

☆ U. S. Government Printing Office: 1945—652512

For sale by the Superintendent of Documents, U. S. Government Printing
Office, Washington 25, D. C.

Transcriber Notes

All illustrations were moved so as to not split paragraphs.

*** END OF THE PROJECT GUTENBERG EBOOK POISON-IVY,
POISON-OAK AND POISON SUMAC ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project GutenbergTM License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.